

# VR IS THE FUTURE OF ENTERTAINMENT



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# **INTERNATIONAL ISLAMIC UNIVERSITY ISLAMABAD**



## **DEPARTMENT OF SOFTWARE ENGINEERING**

### **INTRODUCTION TO SOFTWARE ENGINEERING**

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**1. INTRODUCTION**

**1.1 PURPOSE**

**1.2 SCOPE**

**1.3 WORK BREAKDOWN STRUCTURE**

**2. OVERALL DESCRIPTION**

**2.1 PRODUCT PRESPECTIVE**

**2.2 OPERATING ENVIORNMENT**

**2.3 DESIGN AND IMPLEMENTATION CONSTRAINTS**

**3. REQUIREMENT IDENTIFYING  
TECHNIQUE**

**3.1 USE CASE DIAGRAM**

**3.2 USE CASE DESCRIPTION**

**4. FUNCTIONAL REQUIREMENTS**

**5. NONFUNCTIONAL REQUIREMENTS**

**6. SYSTEM OVERVIEW**

**6.1 ARCHITECTURAL DESIGN**

**7. DESIGN MODELS**

**ALL DIAGRAMS**

**8. TESTING STRATEGIES**

**9. REFERENCES**

# PROJECT PLANNER

| No. of changes | TASK                                       | SUB-TASK                                                                                                                                                                                                  | TIMELINE          |
|----------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1              | <b>Project Initialization and Overview</b> | 1.1. Gather Project Overview - Define project scope<br>1.2. Define Functional and Non-Functional Requirements                                                                                             | 1ST WEEK          |
| 2              | <b>System Design and Modelling</b>         | 2.1. Create Full Dress Use Case<br>2.1.1. Identify Actors - Users, external systems<br>2.1.2. Diagram Use Cases - Use case diagram with relationships (includes, extends)<br>2.3. Create Activity Diagram | 2ND WEEK          |
| 3              | <b>System Design and Modelling 2</b>       | 3.1. Create Class Diagram<br>3.2. Create Data Flow Diagram (DFD)                                                                                                                                          | START OF 3RD WEEK |
| 4              | <b>Documentation and Reporting</b>         | 4.1. Design Documentation - Include use cases, activity diagrams, class diagrams, DFD<br>4.2. Create and Submit Report                                                                                    | END OF 3RD WEEK   |

# VR IS THE FUTURE OF ENTERTAINMENT

## 1. Introduction

### 1.1 Purpose

This document outlines the key requirements, features, and constraints for developing innovative Virtual Reality (VR) glasses. The aim is to create an immersive, reliable, and user-friendly device that serves a variety of applications, such as gaming, education, healthcare, and professional training. It highlights the specifications and guidelines needed to ensure high performance, safety, and a seamless user experience.

### 1.2 Scope

The VR glasses are designed to deliver an immersive virtual experience with portability and ease of use at their core. The product will cater to diverse use cases, focusing on:

- Reducing latency to ensure smooth and responsive interactions.
- Delivering a high refresh rate for realistic visuals and reduced motion sickness.
- Supporting a wide range of applications across different industries.
- Incorporating a durable, lightweight, and ergonomic design for user comfort.
- Meeting stringent safety and environmental standards.

The glasses will work seamlessly with PCs, gaming consoles, and mobile devices, ensuring versatile compatibility and stable connectivity.

### 1.3 Work Breakdown Structure

The development process is divided into several key stages:

1. **Requirement Analysis:** Understanding user needs and technical requirements.
2. **Design:** Developing the overall structure and specifications for hardware and software.
3. **Development:** Building and integrating the hardware and software components.
4. **Testing:** Ensuring the product meets performance, compatibility, and safety benchmarks.
5. **Deployment:** Launching the glasses and providing initial user support.
6. **Maintenance:** Offering regular updates and addressing issues as they arise.

## 2. Overall Description

### 2.1 Product Perspective

The VR glasses are designed to be a versatile and lightweight device that integrates smoothly with external systems, including PCs, consoles, and smartphones. They utilize high-resolution displays and advanced motion-tracking technology to provide an immersive experience.

Compared to traditional VR setups, this product emphasizes portability, comfort, and energy efficiency. It supports a variety of applications, such as gaming, training simulations, and augmented reality (AR) overlays, making it suitable for both personal and professional use.

## 2.2 Operating Environment

The VR glasses will perform effectively in:

- **Indoor Settings:** Ideal for home use, offices, and classrooms.
- **Outdoor Environments:** Equipped to handle temperature variations, light exposure, and humidity.
- **Specialized Workspaces:** Built for use in industrial or professional settings requiring durability and safety compliance.

The glasses will be designed for continuous use, with features to manage comfort, temperature, and external disturbances like noise.

## 2.3 Design and Implementation Constraints

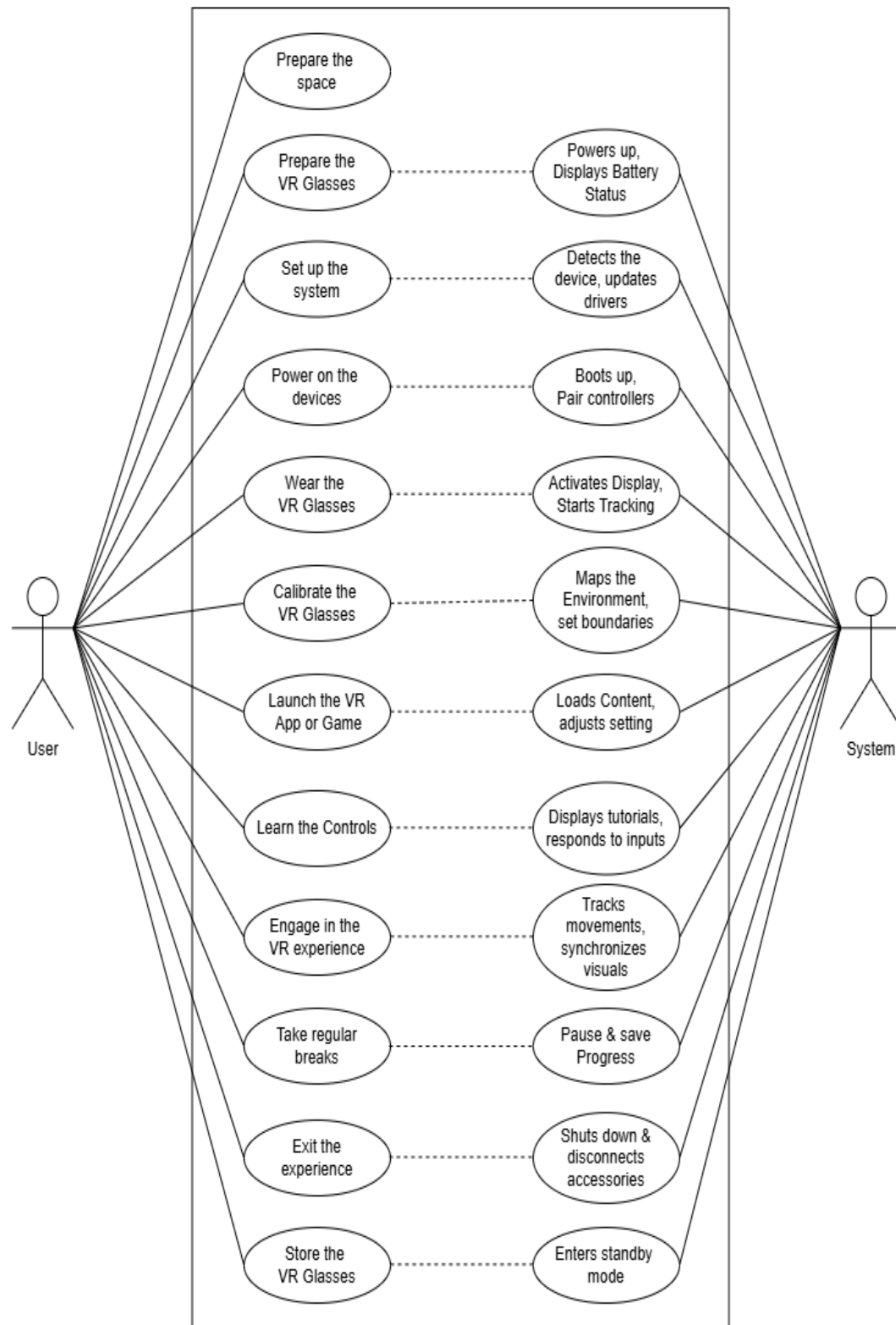
- **Hardware Limitations:**
  - The materials used must balance durability with lightweight construction for long-term comfort.
  - Efficient cooling systems are essential to prevent overheating during extended sessions.
- **Software Requirements:**
  - Real-time responsiveness with minimal latency.
  - Compatibility with various operating systems and app ecosystems.
- **Compliance Standards:**
  - Must adhere to safety regulations for wearable devices and electronic gadgets.
  - Eco-friendly materials and energy-efficient components to meet environmental standards.
- **Power Management:**
  - Battery life must support at least six hours of uninterrupted usage.
  - Components must be optimized for low energy consumption to improve heat management.
- **Connectivity and Integration:**
  - Reliable wireless performance with low latency.
  - Cross-platform compatibility for seamless integration across devices.

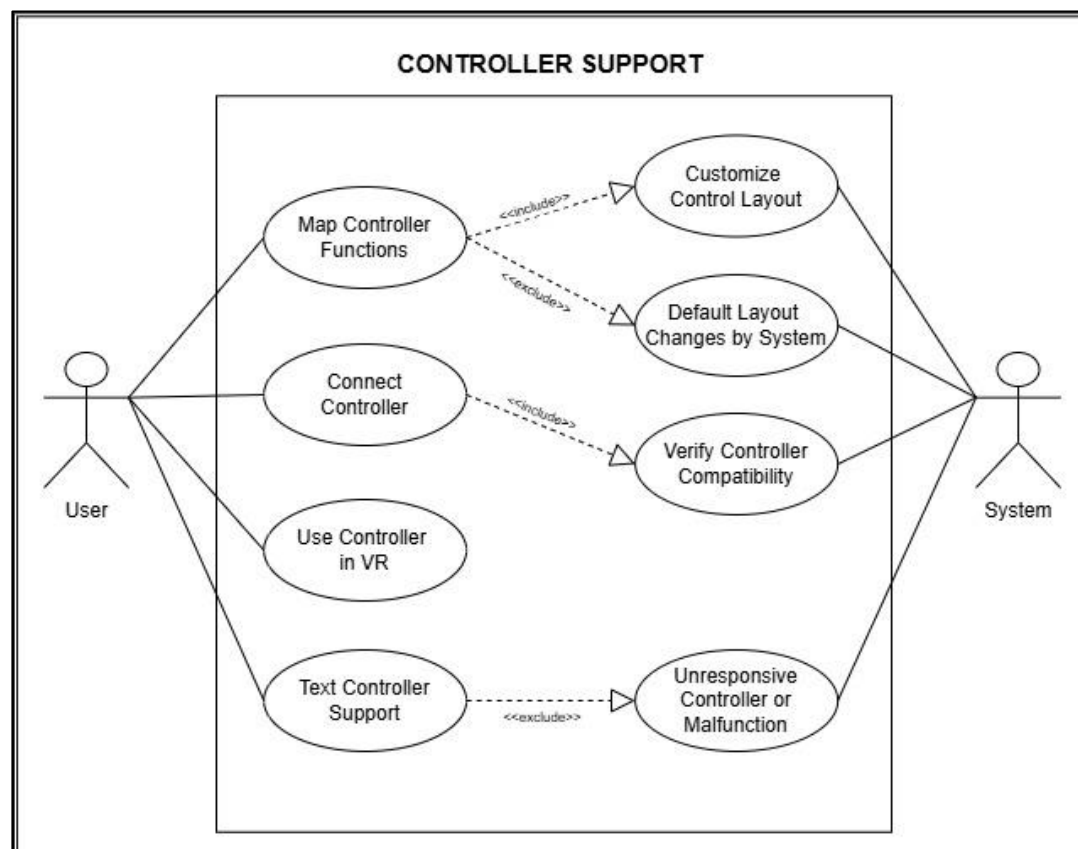
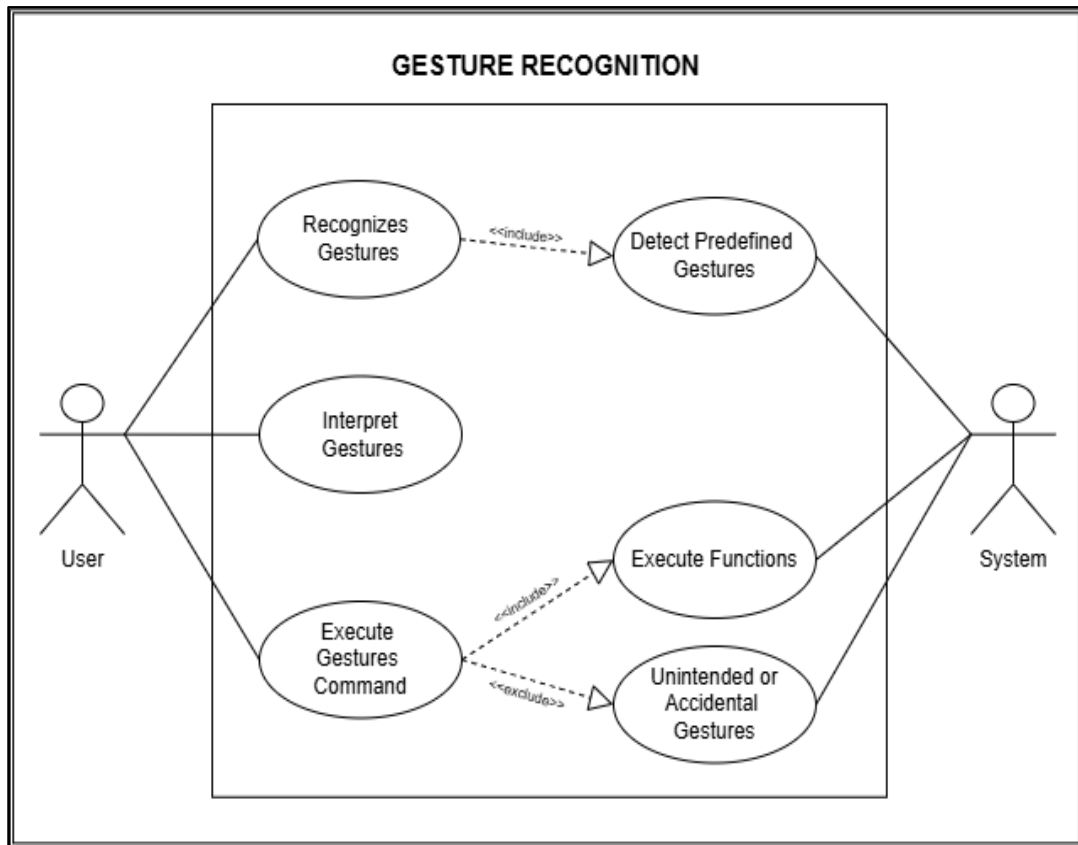
# 3. REQUIREMENT IDENTIFYING TECHNIQUE:

## 3.1 USE CASE DIAGRAM

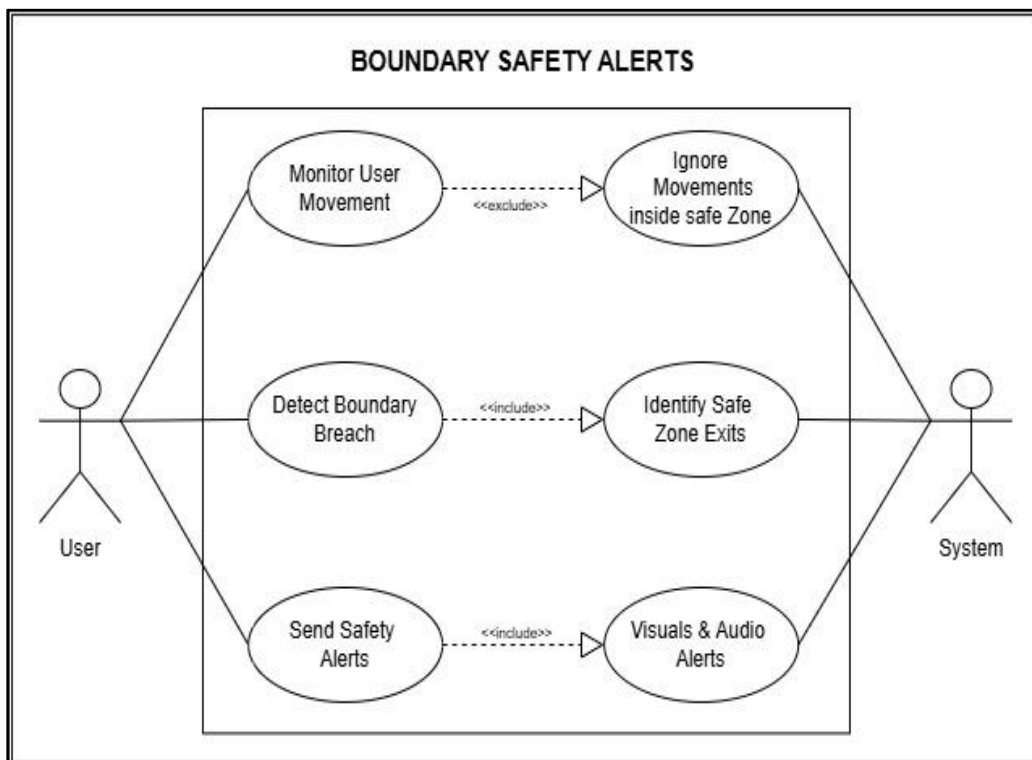
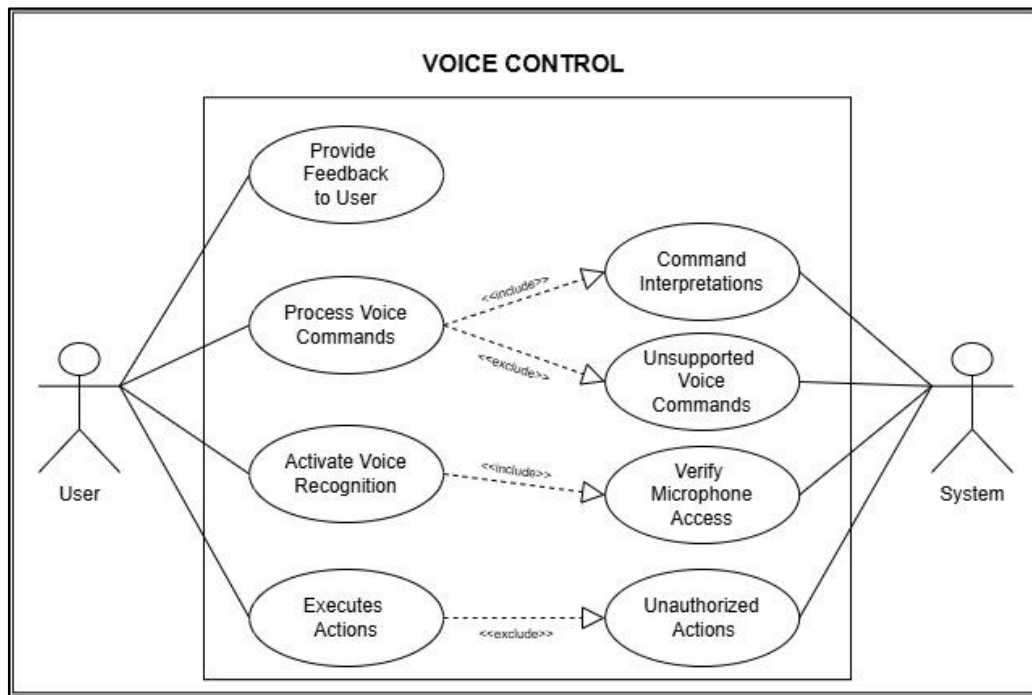
### FULL SYSTEM USE CASE DIAGRAM

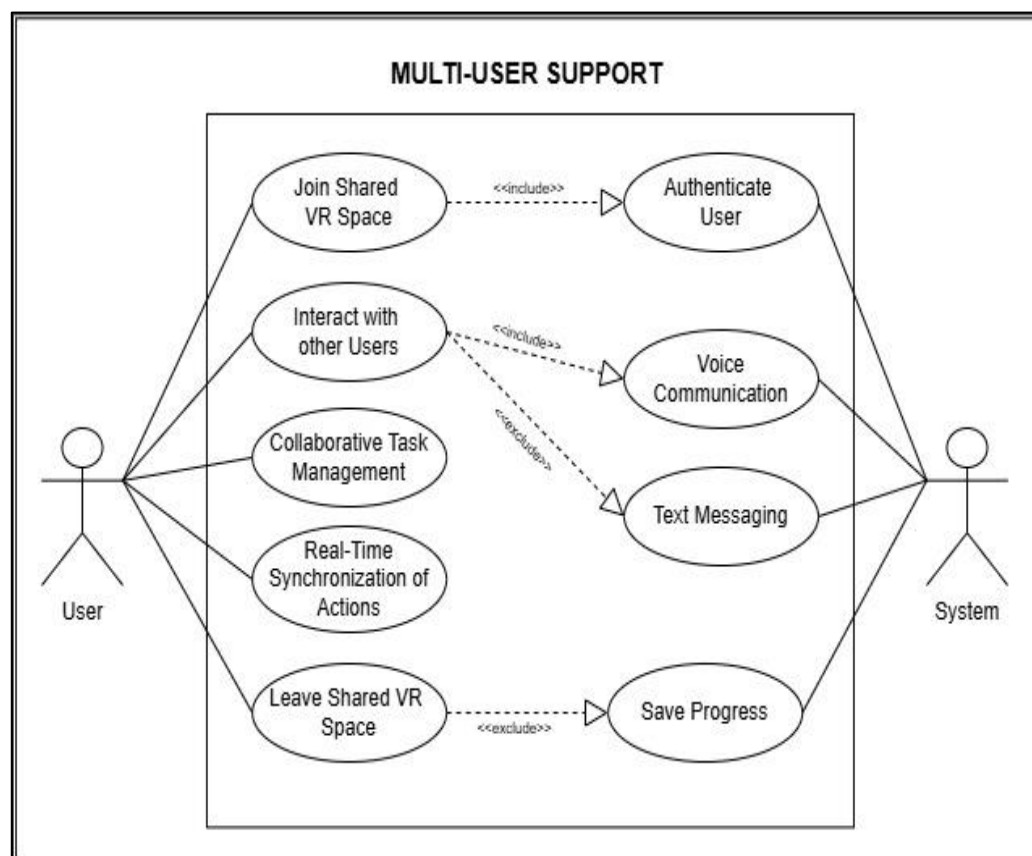
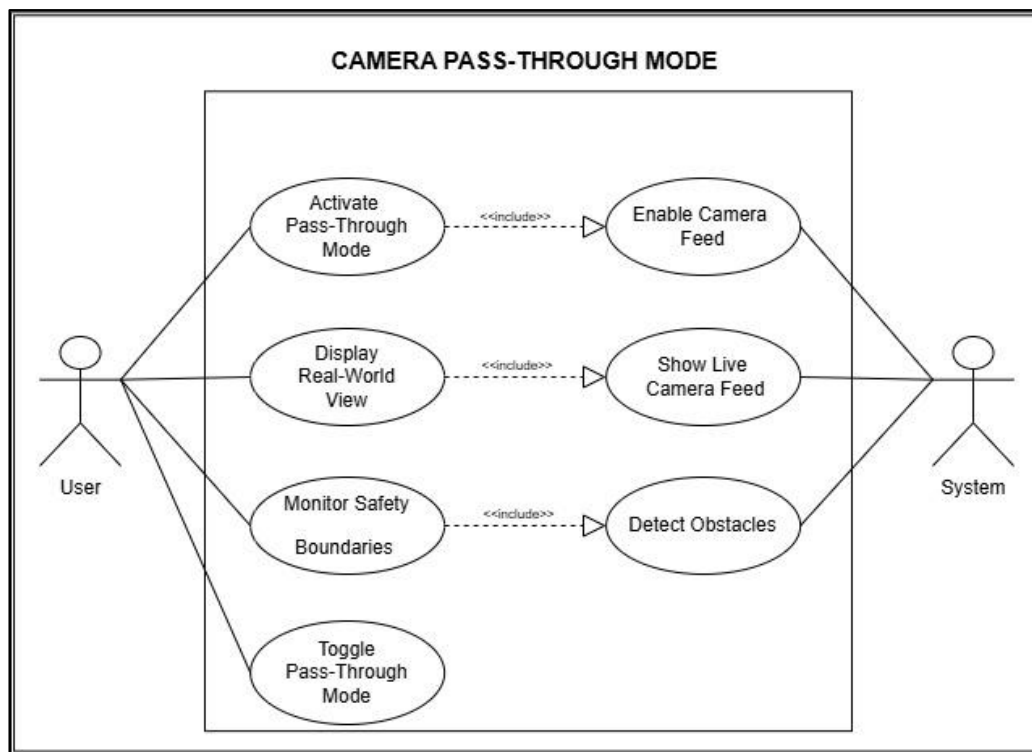
### VR GLASSES USE CASE DIAGRAM

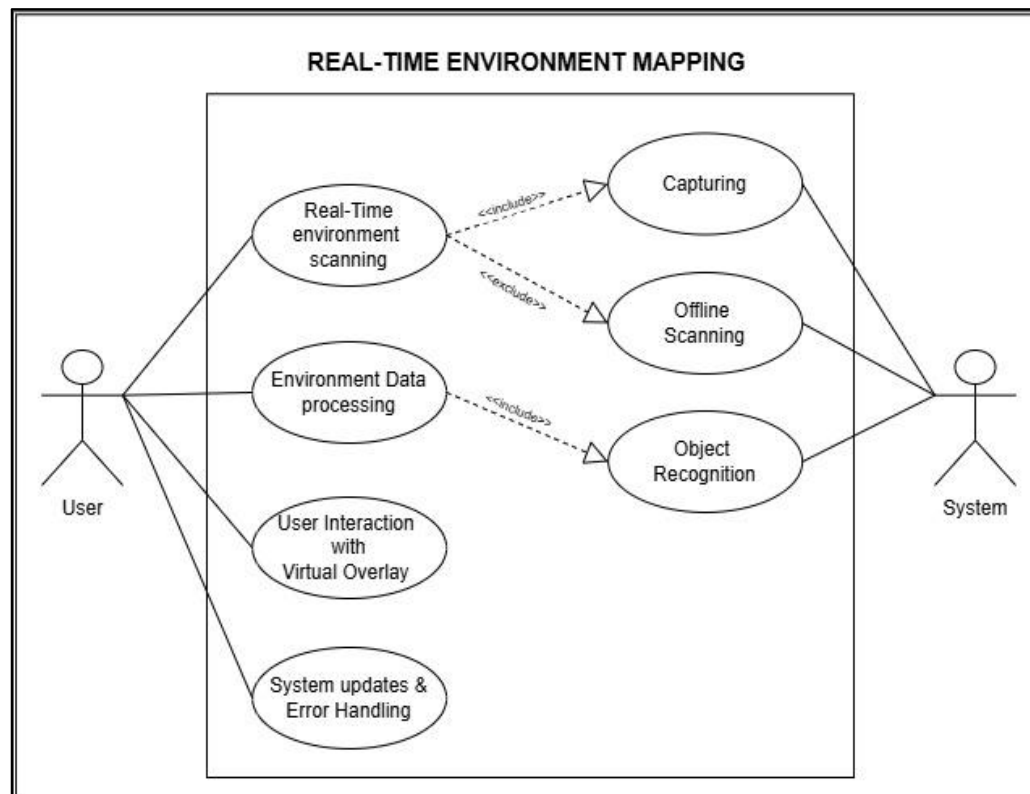
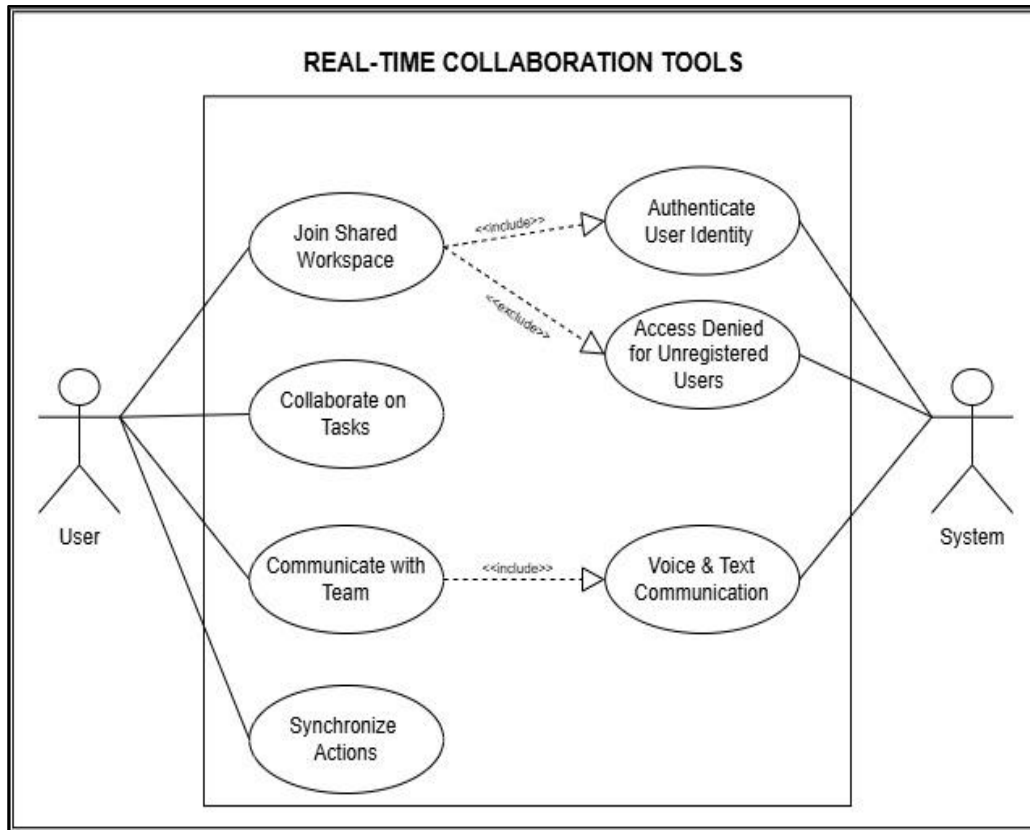


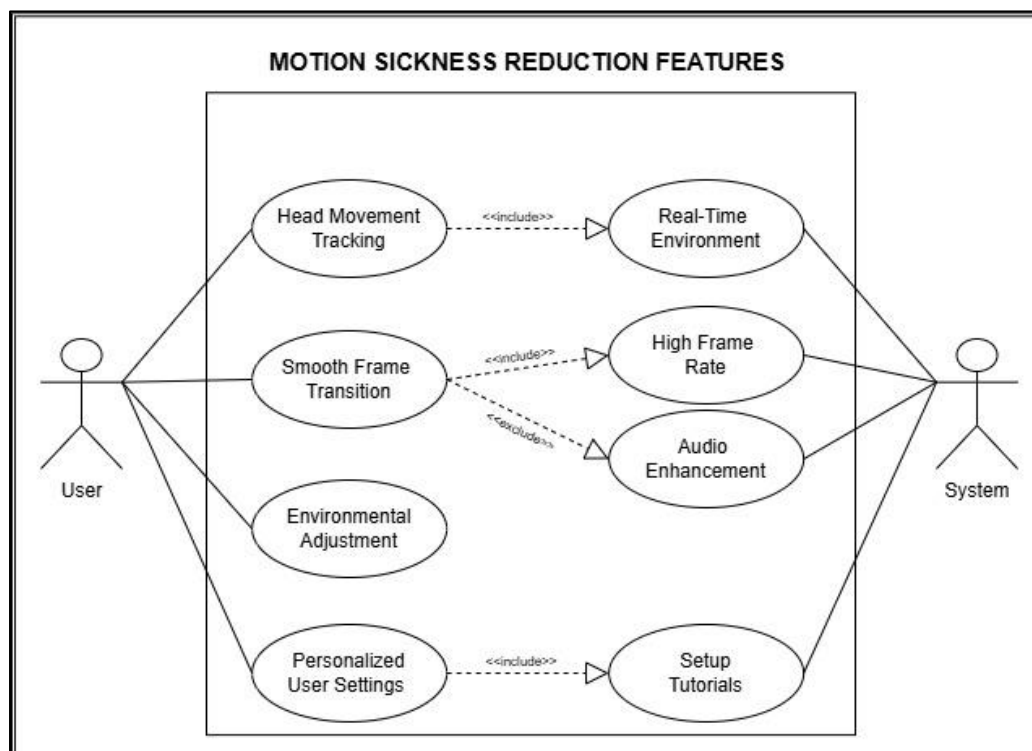
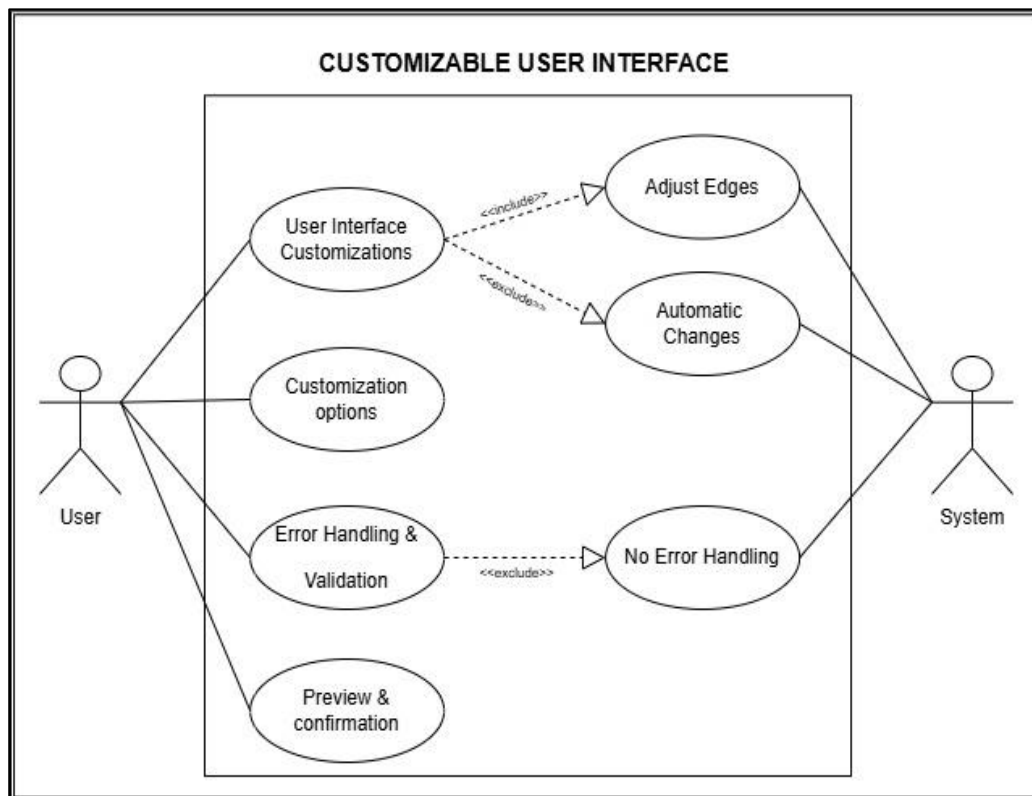












## 3.2 USE CASE DESCRIPTION

| Use case section           | Comment                                                                                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case name              | Gesture Recognition                                                                                                                                                                                                                               |
| Scope                      | Allow users to control the VR environment with hand gestures.                                                                                                                                                                                     |
| Level                      | User-goal level: Focus on easy control through gestures.                                                                                                                                                                                          |
| Primary Actor              | The user performing gestures to control the VR system.                                                                                                                                                                                            |
| Stakeholders and Interests | <ul style="list-style-type: none"><li>• <b>User:</b> Wants easy and accurate gesture control.</li><li>• <b>Developer:</b> Wants the system to work smoothly.</li><li>• <b>Hardware Manufacturer:</b> Wants to improve gesture sensors.</li></ul>  |
| Preconditions              | <ul style="list-style-type: none"><li>• VR glasses and sensors are turned on and connected.</li><li>• There is enough light for the sensors to detect gestures.</li></ul>                                                                         |
| Success Guarantee          | The system correctly recognizes gestures and performs actions.                                                                                                                                                                                    |
| Main success scenario      | <ul style="list-style-type: none"><li>• The user wears the VR glasses.</li><li>• The user makes a gesture.</li><li>• The system detects the gesture and performs the action.</li></ul>                                                            |
| Extensions                 | <ul style="list-style-type: none"><li>• 2a. Gesture not recognized: The system asks the user to try again.</li><li>• 3a. Too dark: The system asks the user to adjust the lighting.</li></ul>                                                     |
| Special requirements       | <ul style="list-style-type: none"><li>• Gestures must be recognized with at least 95% accuracy.</li><li>• System should respond in less than 200 milliseconds.</li><li>• Must be easy for all users, including those with disabilities.</li></ul> |
|                            |                                                                                                                                                                                                                                                   |

| Use case section           | Comment                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case name              | Controller Support                                                                                                                                                                                                                                                                 |
| Scope                      | Allow the VR glasses to work smoothly with a handheld controller for easy use.                                                                                                                                                                                                     |
| Level                      | User-goal level: Focuses on the end-user's ability to use the controller effectively with the VR glasses.                                                                                                                                                                          |
| Primary Actor              | VR user                                                                                                                                                                                                                                                                            |
| Stakeholders and Interests | <ul style="list-style-type: none"> <li>• <b>Users:</b> Want easy and responsive controls.</li> <li>• <b>Manufacturers:</b> Need the controller and VR glasses to work well together.</li> <li>• <b>Developers:</b> Need reliable controller performance for their apps.</li> </ul> |
| Preconditions              | The VR glasses and controller are turned on, charged, and connected.                                                                                                                                                                                                               |
| Success Guarantee          | The controller works perfectly, letting the user navigate VR without problems.                                                                                                                                                                                                     |
| Main success scenario      | <ul style="list-style-type: none"> <li>• User turns on the VR glasses and controller.</li> <li>• They connect to each other.</li> <li>• User moves and interacts in VR without issues.</li> </ul>                                                                                  |
| Extensions                 | <ul style="list-style-type: none"> <li>• If pairing fails, the system shows how to fix it.</li> <li>• If the controller battery is low, it warns the user.</li> </ul>                                                                                                              |
| Special requirements       | The controller should vibrate for feedback and be comfortable to use for a long time.                                                                                                                                                                                              |

| Use case section | Comment                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------|
| Use case name    | Voice Support                                                                                 |
| Scope            | Use voice commands to control VR glasses without using hands.                                 |
| Level            | User-goal level: Focuses on the user's ability to control VR applications via voice commands. |

|                                   |                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor</b>              | The person using the VR glasses.                                                                                                                                                                                     |
| <b>Stakeholders and Interests</b> | <ul style="list-style-type: none"> <li>• <b>User:</b> Easy and smooth control.</li> <li>• <b>Manufacturer:</b> More features to attract customers.</li> <li>• <b>Developers:</b> Ensure it works reliably</li> </ul> |
| <b>Preconditions</b>              | The VR glasses are on, and voice control is ready.                                                                                                                                                                   |
| <b>Success Guarantee</b>          | The glasses respond correctly to voice commands and give feedback                                                                                                                                                    |
| <b>Main success scenario</b>      | <ul style="list-style-type: none"> <li>• User turns on voice control.</li> <li>• Says a command (e.g., "Open Menu").</li> <li>• The glasses understand and follow the command.</li> </ul>                            |
| <b>Extensions</b>                 | <ul style="list-style-type: none"> <li>• If the command is unclear, the system asks the user to repeat.</li> <li>• If the system doesn't hear, it alerts the user.</li> </ul>                                        |
| <b>Special requirements</b>       | should work in different languages, noisy places, and be easy for everyone to use.                                                                                                                                   |

| <b>Use case section</b>           | <b>Comment</b>                                                                                                                                                                         |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case name</b>              | Boundary safety alerts                                                                                                                                                                 |
| <b>Scope</b>                      | The system helps users stay safe by warning them when they get too close to real-world obstacles while using VR glasses.                                                               |
| <b>Level</b>                      | Basic user goals.                                                                                                                                                                      |
| <b>Primary Actor</b>              | The person using the VR glasses.                                                                                                                                                       |
| <b>Stakeholders and Interests</b> | <ul style="list-style-type: none"> <li>• <b>User:</b> Wants to stay safe and avoid hitting objects.</li> <li>• <b>VR Maker:</b> Wants to make the product safe and reliable</li> </ul> |
| <b>Preconditions</b>              | The VR glasses need to know the room's boundaries.                                                                                                                                     |

|                              |                                                                                                                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Success Guarantee</b>     | The user gets a warning in time to avoid danger or damage.                                                                                                                                                                                                                          |
| <b>Main success scenario</b> | <ul style="list-style-type: none"> <li>• The user is playing or working in VR.</li> <li>• The glasses notice the user is close to a wall or object.</li> <li>• A warning (like a sound, light, or vibration) is sent.</li> <li>• The user stops or moves back to safety.</li> </ul> |
| <b>Extensions</b>            | <ul style="list-style-type: none"> <li>• If the glasses can't detect boundaries, they ask the user to reset them.</li> <li>• If the sound is hard to hear, the glasses vibrate instead.</li> </ul>                                                                                  |
| <b>Special requirements</b>  | <ul style="list-style-type: none"> <li>• Warnings must be clear but not distracting.</li> <li>• The system should work even in dark or cluttered spaces.</li> </ul>                                                                                                                 |

| <b>Use case section</b>           | <b>Comment</b>                                                                                                                                                                                                                               |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case name</b>              | Camera Pass-Through Mode                                                                                                                                                                                                                     |
| <b>Scope</b>                      | Enables the user to see the real-world environment through the VR glasses using the camera.                                                                                                                                                  |
| <b>Level</b>                      | The user can switch between virtual reality and the real-world view when necessary.                                                                                                                                                          |
| <b>Primary Actor</b>              | VR user.                                                                                                                                                                                                                                     |
| <b>Stakeholders and Interests</b> | <p>Users want to see the physical environment for safety and convenience.</p> <p>Developers ensure smooth, high-quality camera feed for real-world visibility.</p> <p>Testers check the system for minimal latency and accurate display.</p> |
| <b>Preconditions</b>              | <p>The system is powered on and the camera is functional.</p> <p>The user is in VR and has the option to switch to pass-through mode.</p>                                                                                                    |



|                              |                                                                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Success Guarantee</b>     | The user can seamlessly switch between virtual and real-world views with minimal delay.                                                                                                                                                                           |
| <b>Main success scenario</b> | <ol style="list-style-type: none"> <li>1. The user activates pass-through mode from the menu.</li> <li>2. The system displays the real-time camera feed.</li> <li>3. The user can interact with the real world through the camera, with minimal delay.</li> </ol> |
| <b>Extensions</b>            | <p>If the camera fails, the system reverts to the VR environment.</p> <p>The system switches to low-light mode automatically in dark conditions.</p>                                                                                                              |
| <b>Special requirements</b>  | <p>Real-time camera feed with low latency.</p> <p>High-definition quality for accurate pass-through views.</p>                                                                                                                                                    |

| <b>Use case section</b>           | <b>Comment</b>                                                                                                                                                                                                                                                   |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case name</b>              | Multi-User Support                                                                                                                                                                                                                                               |
| <b>Scope</b>                      | Enables multiple users to interact and collaborate in the same VR space.                                                                                                                                                                                         |
| <b>Level</b>                      | VR user                                                                                                                                                                                                                                                          |
| <b>Primary Actor</b>              | User: The user joins a shared VR space and collaborates or interacts with other users, needing real-time synchronization and communication.                                                                                                                      |
| <b>Stakeholders and Interests</b> | <p>Users want to interact seamlessly with others and synchronize actions in real-time.</p> <p>Developers ensure smooth communication and synchronization among users.</p> <p>Testers verify that the system can handle multiple users without lag or errors.</p> |
| <b>Preconditions</b>              | <p>All users are logged into the shared VR space with compatible hardware.</p> <p>The system supports multiple users interacting in real-time.</p>                                                                                                               |
| <b>Success Guarantee</b>          | All users can communicate, synchronize actions, and interact smoothly with one another.                                                                                                                                                                          |
| <b>Main success scenario</b>      | <ol style="list-style-type: none"> <li>1. The user enters a shared VR space with other users.</li> </ol>                                                                                                                                                         |

|                             |                                                                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>2. The system synchronizes movements, actions, and communication in real-time.</p> <p>3. The user can collaborate, communicate, and interact with others without delay.</p> |
| <b>Extensions</b>           | Network lag causes delayed actions, and the system displays a warning. Users experience issues with communication tools due to connectivity problems.                          |
| <b>Special requirements</b> | <p>Real-time synchronization with low-latency performance.</p> <p>Reliable communication tools (voice, gestures) for smooth interaction.</p>                                   |

| <b>Use case section</b>           | <b>Comment</b>                                                   |
|-----------------------------------|------------------------------------------------------------------|
| <b>Use case name</b>              | Real-Time Environment Mapping                                    |
| <b>Scope</b>                      | Maps and adapts to the user's physical environment in real time. |
| <b>Level</b>                      | System feature.                                                  |
| <b>Primary Actor</b>              | VR user                                                          |
| <b>Stakeholders and Interests</b> | Users (safety and immersion)<br>developers (flexibility).        |
| <b>Preconditions</b>              | Sensors and mapping software are active.                         |
| <b>Success Guarantee</b>          | Accurate real-time mapping of the physical environment.          |
| <b>Main success scenario</b>      | Users navigate safely while immersed in VR.                      |
| <b>Extensions</b>                 | AR features to display mapped surroundings.                      |
| <b>Special requirements</b>       | Fast and accurate spatial tracking.                              |

| Use case section           | Comment                                                             |
|----------------------------|---------------------------------------------------------------------|
| Use case name              | Customizable User Interface                                         |
| Scope                      | Usability feature.                                                  |
| Level                      | System feature.                                                     |
| Primary Actor              | VR user                                                             |
| Stakeholders and Interests | Users (tailored experiences)<br>Developers (enhanced engagement).   |
| Preconditions              | Default interface is active, and customization options are enabled. |
| Success Guarantee          | Users can modify settings and layouts easily.                       |
| Main success scenario      | Interface adapts to user preferences.                               |
| Extensions                 | Advanced themes or accessibility adjustments.                       |
| Special requirements       | Intuitive customization tools.                                      |

| Use case section           | Comment                                                             |
|----------------------------|---------------------------------------------------------------------|
| Use case name              | Real Time Collaboration tools                                       |
| Scope                      | Facilitates teamwork and communication in virtual environments.     |
| Level                      | System feature.                                                     |
| Primary Actor              | VR users collaborating in the same session.                         |
| Stakeholders and Interests | Users (effective collaboration), businesses (productivity).         |
| Preconditions              | Users are connected to the same session with communication enabled. |
| Success Guarantee          | Collaboration tools function seamlessly.                            |
| Main success scenario      | Users work or communicate effectively in a shared VR space.         |
| Extensions                 | Support for file sharing or virtual whiteboards.                    |
| Special requirements       | Low-latency communication and secure data sharing.                  |

| Use case section           | Comment                                                        |
|----------------------------|----------------------------------------------------------------|
| Use case name              | Motion Sickness Reduction Features                             |
| Scope                      | Minimizes discomfort during prolonged VR use.                  |
| Level                      | Usability and health feature.                                  |
| Primary Actor              | VR user.                                                       |
| Stakeholders and Interests | Users (comfortable experience), healthcare advocates.          |
| Preconditions              | VR environment is active and optimized for performance.        |
| Success Guarantee          | Reduced motion sickness symptoms for most users.               |
| Main success scenario      | Users work or communicate effectively in a shared VR space.    |
| Extensions                 | User-adjustable comfort settings.                              |
| Special requirements       | High refresh rates, low latency, and adaptive motion handling. |

## 4. FUNCTIONAL REQUIREMENTS:

- 1. High-Resolution Display:** Provide immersive 3D visuals with a high-resolution screen for clear and sharp images. This ensures a lifelike virtual environment with fine details.
- 2. Head Tracking:** Track head movements in real-time for an accurate and responsive virtual experience. This allows users to look around in the virtual world naturally.
- 3. Six Degrees of Freedom (6DoF):** Support full spatial movement, enabling users to move in any direction in 3D space. This enhances interaction with the virtual world, making it more immersive.
- 4. Gesture Recognition:** Enable detection of hand gestures for intuitive interaction without controllers. This allows for touch-free, natural control within the VR environment.
- 5. Controller Support:** Ensure compatibility with external VR controllers for precise navigation and object manipulation. This enhances control during interactive experiences.
- 6. Voice Control:** Integrate voice recognition to allow users to give commands or interact with the VR environment using voice. This facilitates hands-free navigation and accessibility.
- 7. Boundary Safety Alerts:** Warn users when they approach real-world boundaries to avoid physical collisions. This ensures user safety while they are immersed in the virtual world.

8. **Spatial Audio:** Provide 3D spatial audio that matches the movement and position of virtual objects. This enhances immersion by making audio feel directional and realistic.
9. **Camera Pass-Through Mode:** Allow users to see the real world through the camera when needed. This feature helps maintain situational awareness while using the VR glasses.
10. **Multi-User Support:** Enable multiple users to interact within the same VR space. This fosters social interaction and collaborative experiences in shared virtual environments.
11. **Real-Time Environment Mapping:** Scan the real-world environment for a more accurate and dynamic interaction with the space. This allows for better integration of virtual objects into the physical world.
12. **Customizable User Interface:** Allow users to personalize the virtual environment and interface settings. This gives users control over how they interact with the VR experience.
13. **Multi-Platform Compatibility:** Ensure the VR glasses can be used with a wide range of devices and platforms. This enhances versatility and appeal to a broader audience.
14. **Real-Time Collaboration Tools:** Provide tools for collaboration in virtual environments, such as shared documents or virtual whiteboards. This supports teamwork and productivity in VR.
15. **Motion Sickness Reduction Features:** Incorporate design elements that minimize VR-induced motion sickness. This improves comfort and usability during extended sessions.

## 5. NON-FUNCTIONAL REQUIREMENTS:

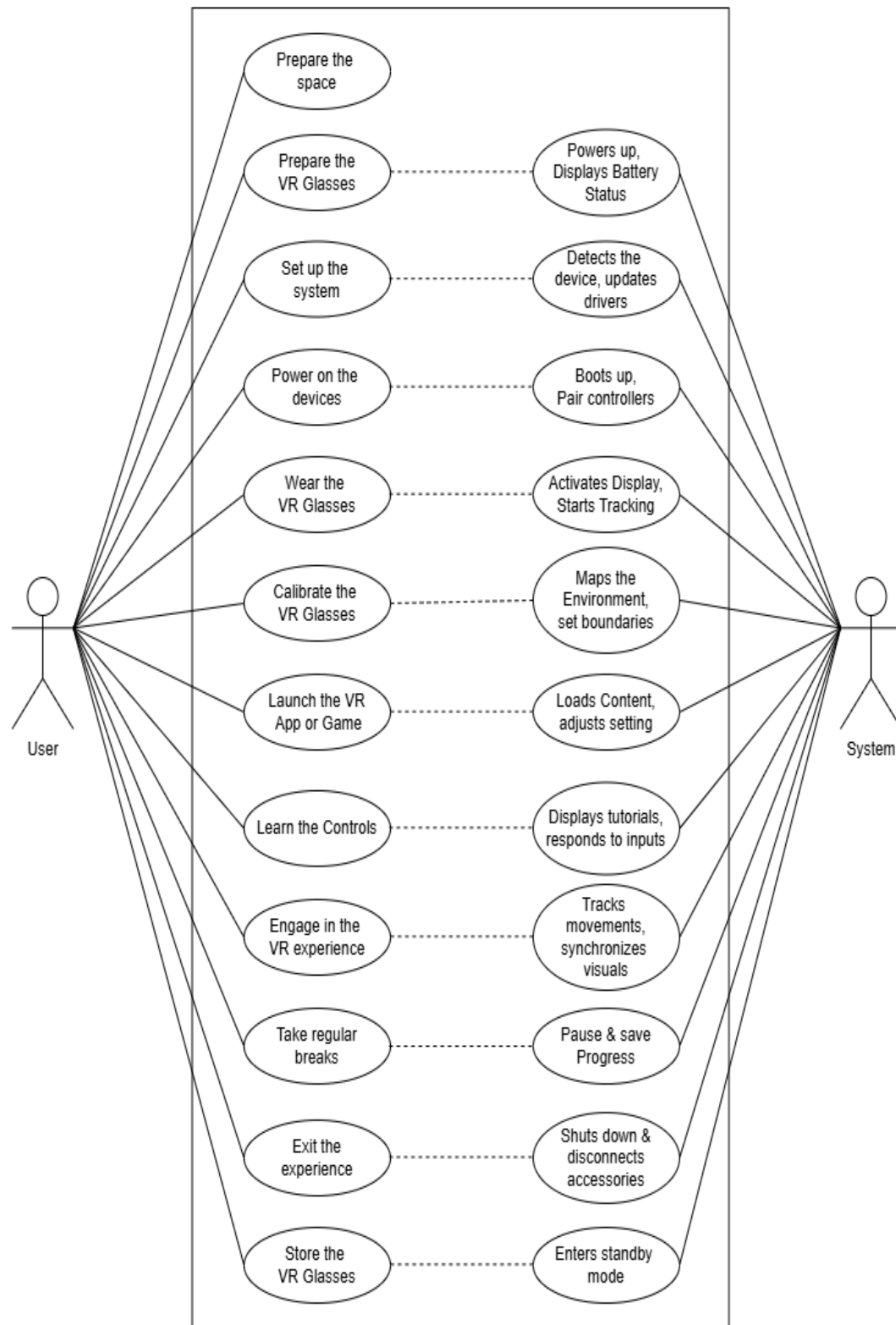
1. **Low Latency:** Ensure minimal delay between user input and visual output. This is crucial to provide a smooth and responsive experience, avoiding motion sickness.
2. **High Refresh Rate:** Maintain a refresh rate of at least 90Hz for fluid visuals. This reduces motion blur and provides a more comfortable, immersive experience.
3. **Lightweight Design:** The device must be lightweight and comfortable to wear for extended periods. This prevents user fatigue and makes the glasses suitable for long sessions.
4. **Usability:** The VR glasses should have a simple setup and an intuitive interface, ensuring ease of use for both beginners and advanced users.
5. **Reliability:** The VR glasses should perform consistently without crashing or malfunctioning. This ensures that users can rely on the device for uninterrupted sessions.
6. **Data Privacy:** All user data must be securely stored and protected, ensuring compliance with privacy regulations. This includes any sensitive data collected during use.
7. **Energy Efficiency:** Optimize power consumption to extend battery life, allowing for longer VR sessions without frequent recharging.
8. **Durability:** The glasses should be resistant to wear and tear, using high-quality materials to withstand drops and regular use.

9. **Efficient Cooling:** The device must have an efficient cooling system to prevent overheating during extended VR use. This ensures consistent performance and comfort.
10. **Compatibility with New Apps:** The system should be easily updated to support new applications. This ensures that the device remains functional and relevant as new VR experiences are developed.
11. **Safety Standards Compliance:** The glasses must meet international safety standards. This ensures that the device is safe for consumer use, preventing health or injury risks.
12. **Comfortable Fit:** The design should provide a comfortable, adjustable fit for different head shapes and sizes. This allows for extended use without discomfort.
13. **Robust Connectivity:** Ensure stable and high-speed wireless connectivity for seamless data transmission and smooth VR interactions. This is essential for multiplayer experiences and app downloads.
14. **Environmental Resistance:** The VR glasses should operate effectively in various environmental conditions, including different lighting and temperatures.
15. **Ergonomic Design:** The glasses should have an ergonomic design that distributes weight evenly across the head. This prevents strain and enhances comfort during use.

## 6. SYSTEM OVERVIEW:

### 6.1 ARCHITECTURAL DESIGN

### VR GLASSES USE CASE DIAGRAM

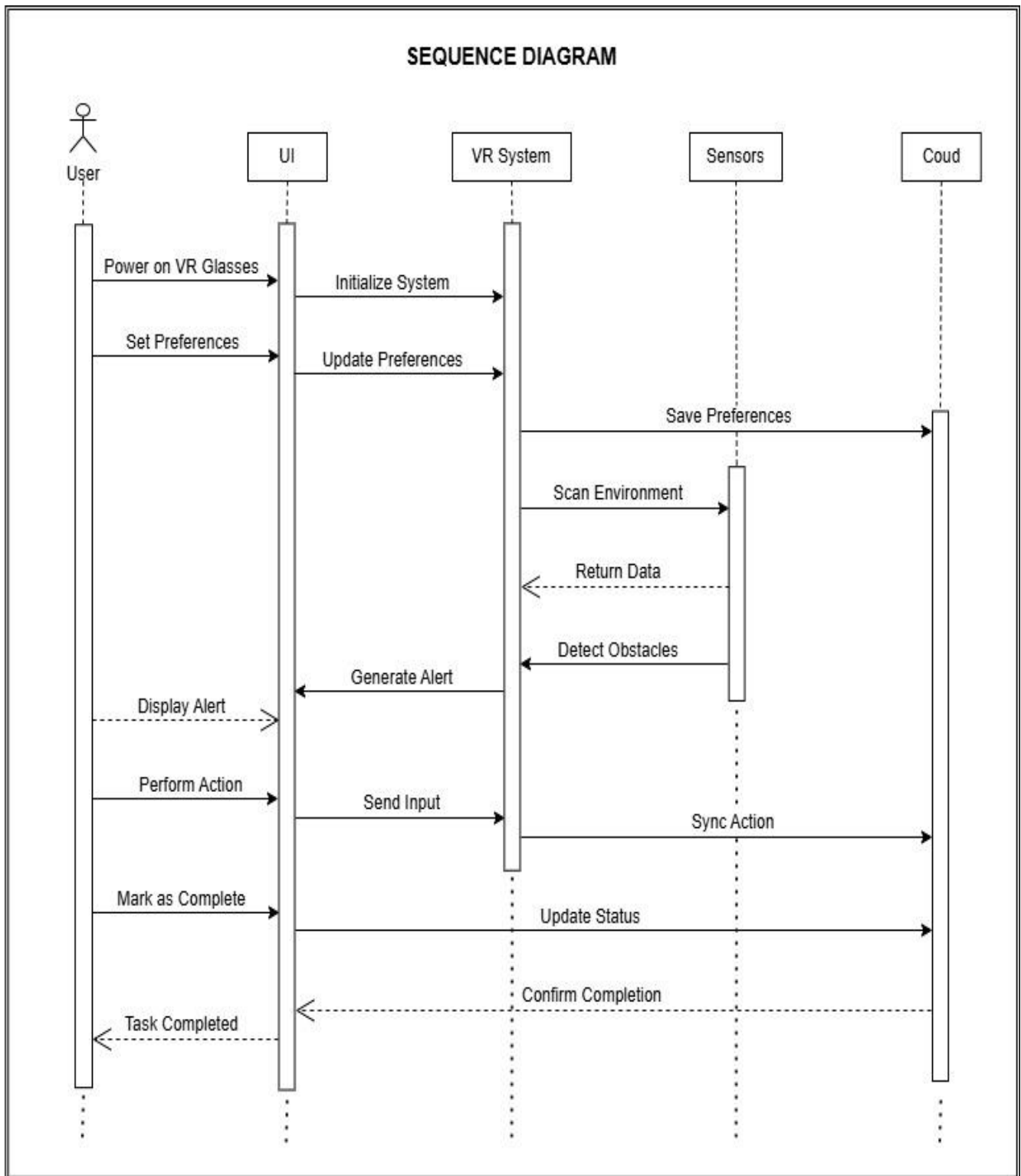


| Use case section           | Comment                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case name              | Controller Support                                                                                                                                                                                                                                                                 |
| Scope                      | Allow the VR glasses to work smoothly with a handheld controller for easy use.                                                                                                                                                                                                     |
| Level                      | User-goal level: Focuses on the end-user's ability to use the controller effectively with the VR glasses.                                                                                                                                                                          |
| Primary Actor              | VR user                                                                                                                                                                                                                                                                            |
| Stakeholders and Interests | <ul style="list-style-type: none"> <li>• <b>Users:</b> Want easy and responsive controls.</li> <li>• <b>Manufacturers:</b> Need the controller and VR glasses to work well together.</li> <li>• <b>Developers:</b> Need reliable controller performance for their apps.</li> </ul> |
| Preconditions              | The VR glasses and controller are turned on, charged, and connected.                                                                                                                                                                                                               |
| Success Guarantee          | The controller works perfectly, letting the user navigate VR without problems.                                                                                                                                                                                                     |
| Main success scenario      | <ul style="list-style-type: none"> <li>• User turns on the VR glasses and controller.</li> <li>• They connect to each other.</li> <li>• User moves and interacts in VR without issues.</li> </ul>                                                                                  |
| Extensions                 | <ul style="list-style-type: none"> <li>• If pairing fails, the system shows how to fix it.</li> <li>• If the controller battery is low, it warns the user.</li> </ul>                                                                                                              |
| Special requirements       | The controller should vibrate for feedback and be comfortable to use for a long time.                                                                                                                                                                                              |

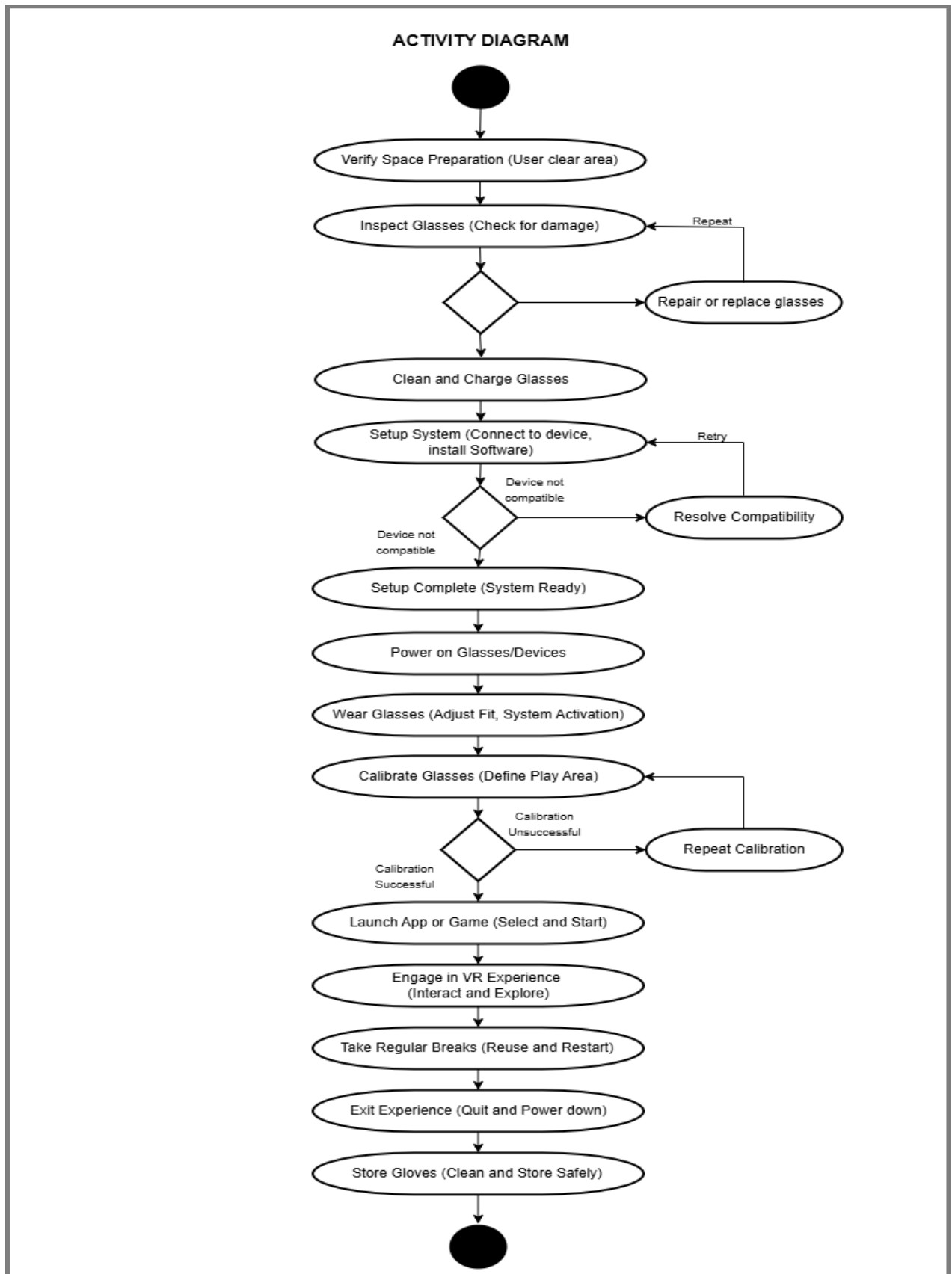


## 7. DESIGN MODELS

### SEQUENCE DIAGRAM:

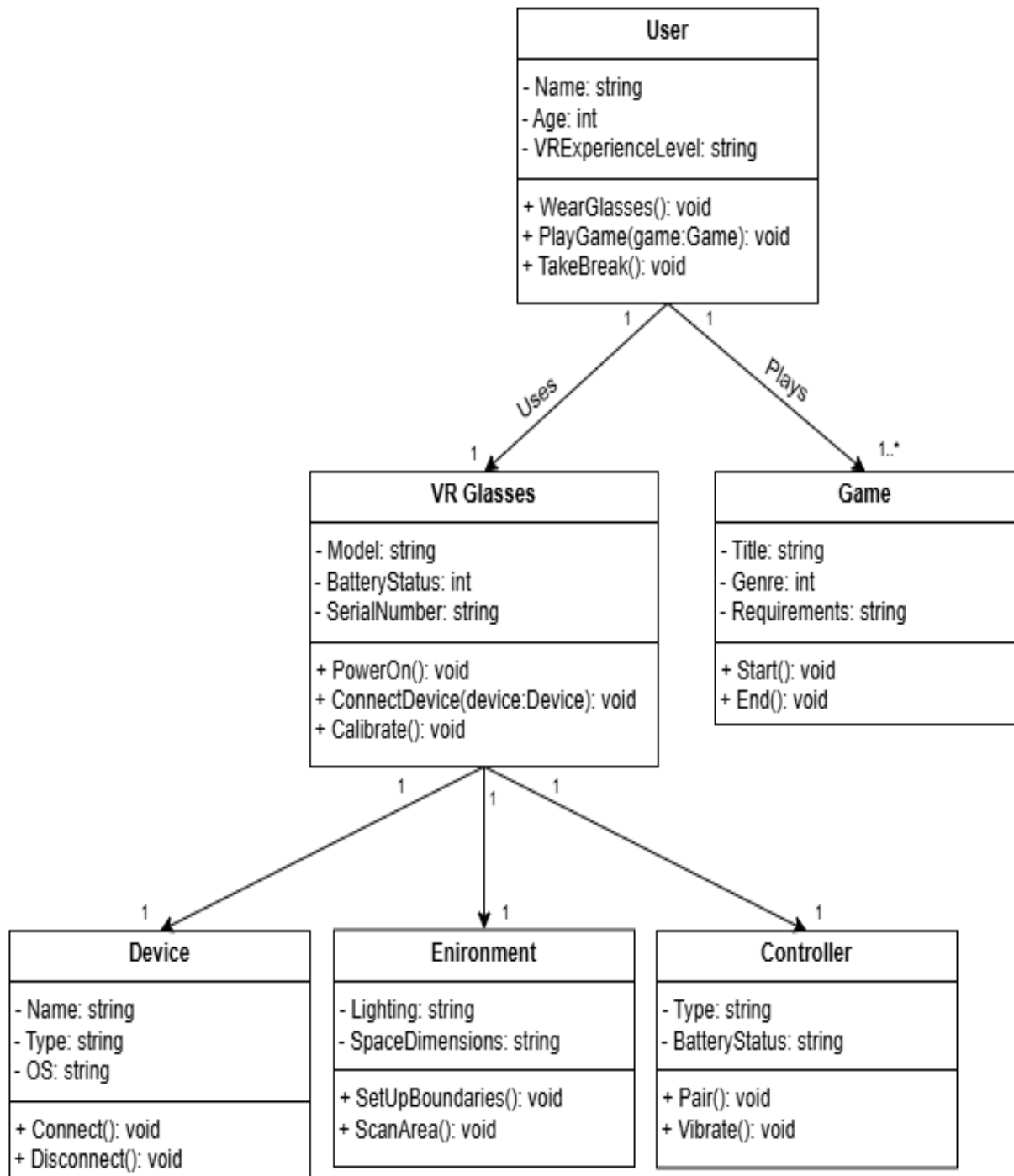


## ACTIVITY DIAGRAM:

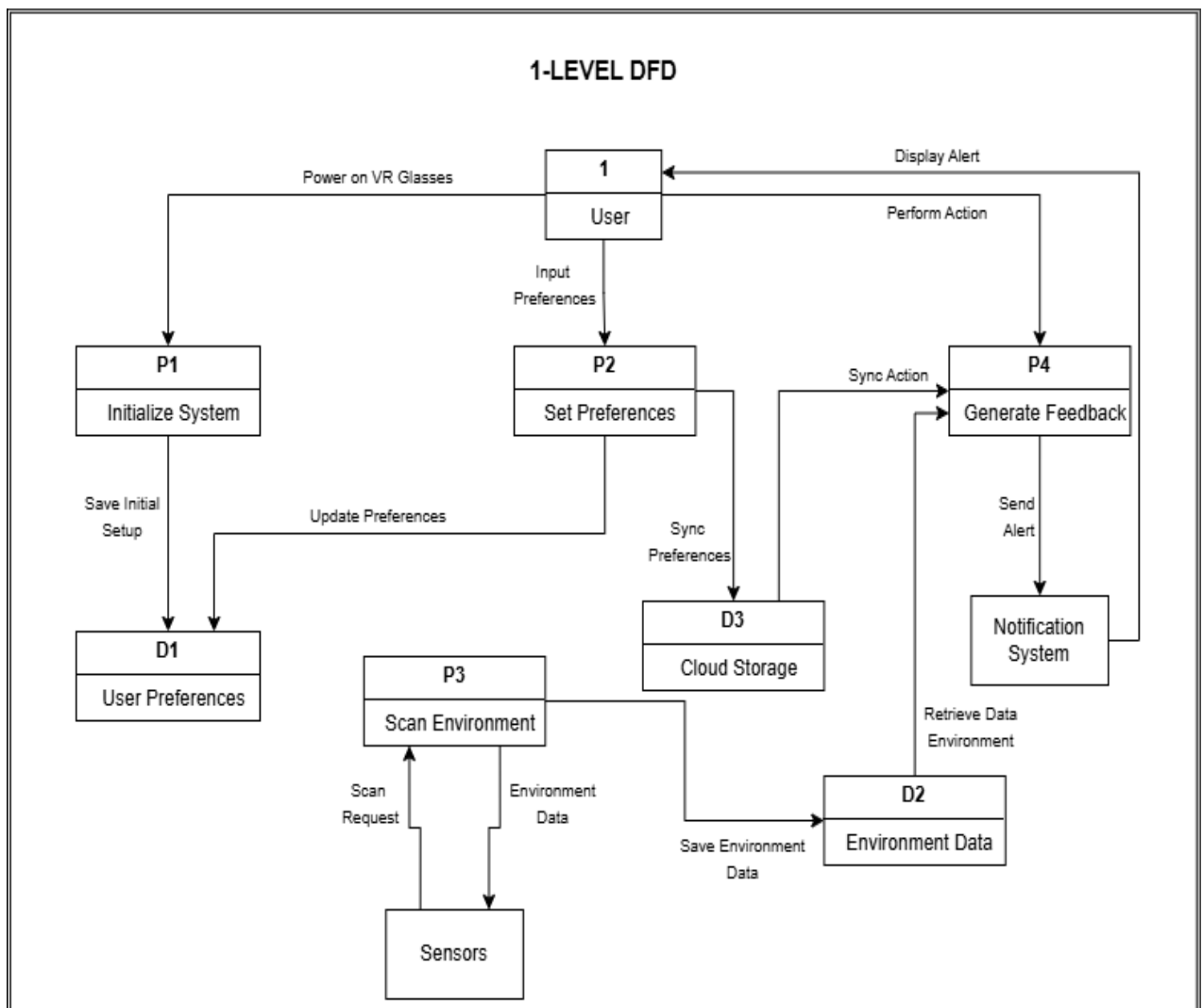
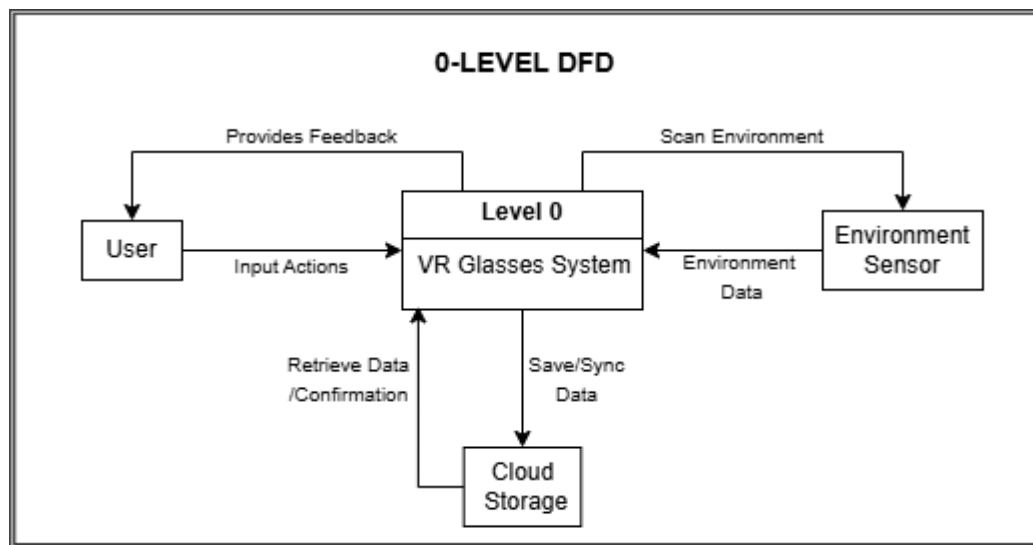


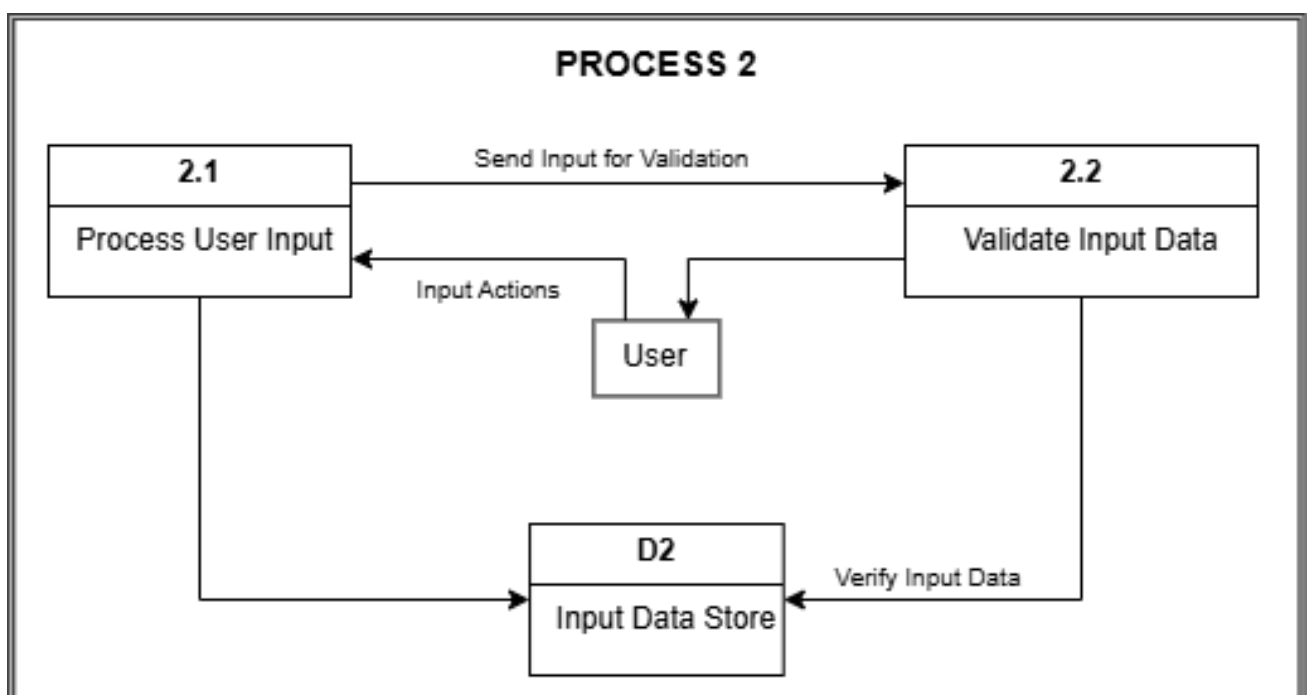
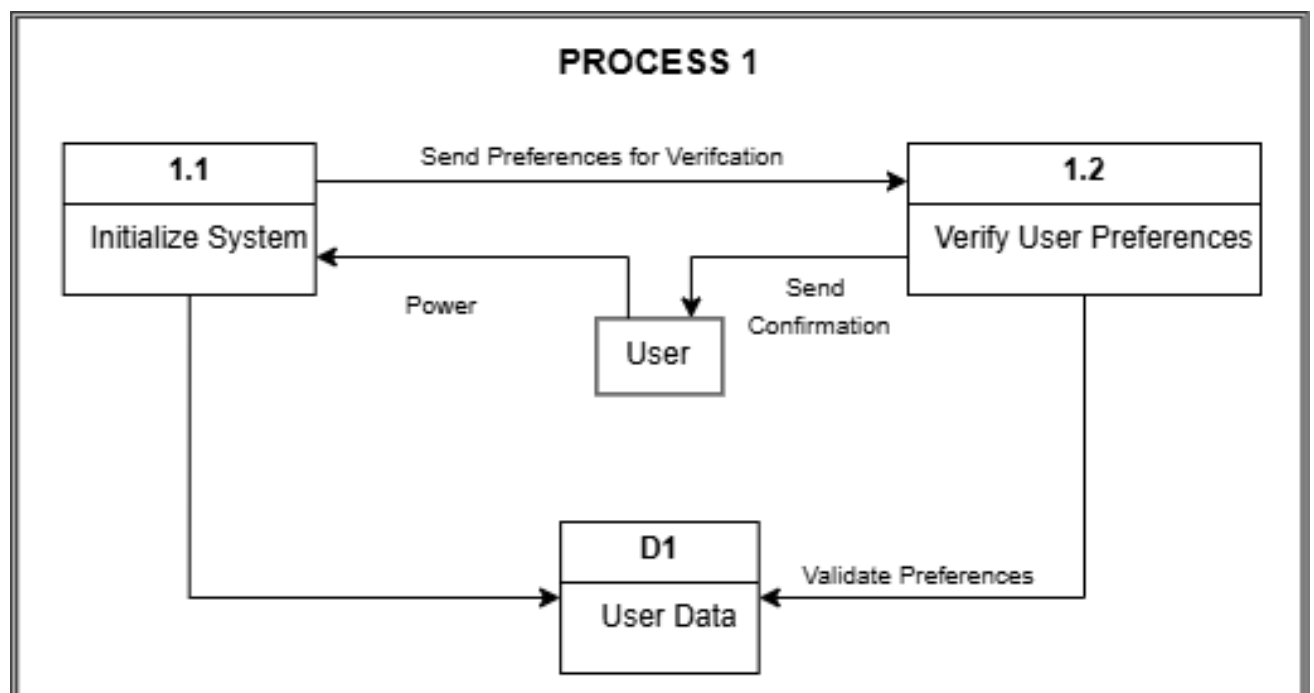
## CLASS DIAGRAM:

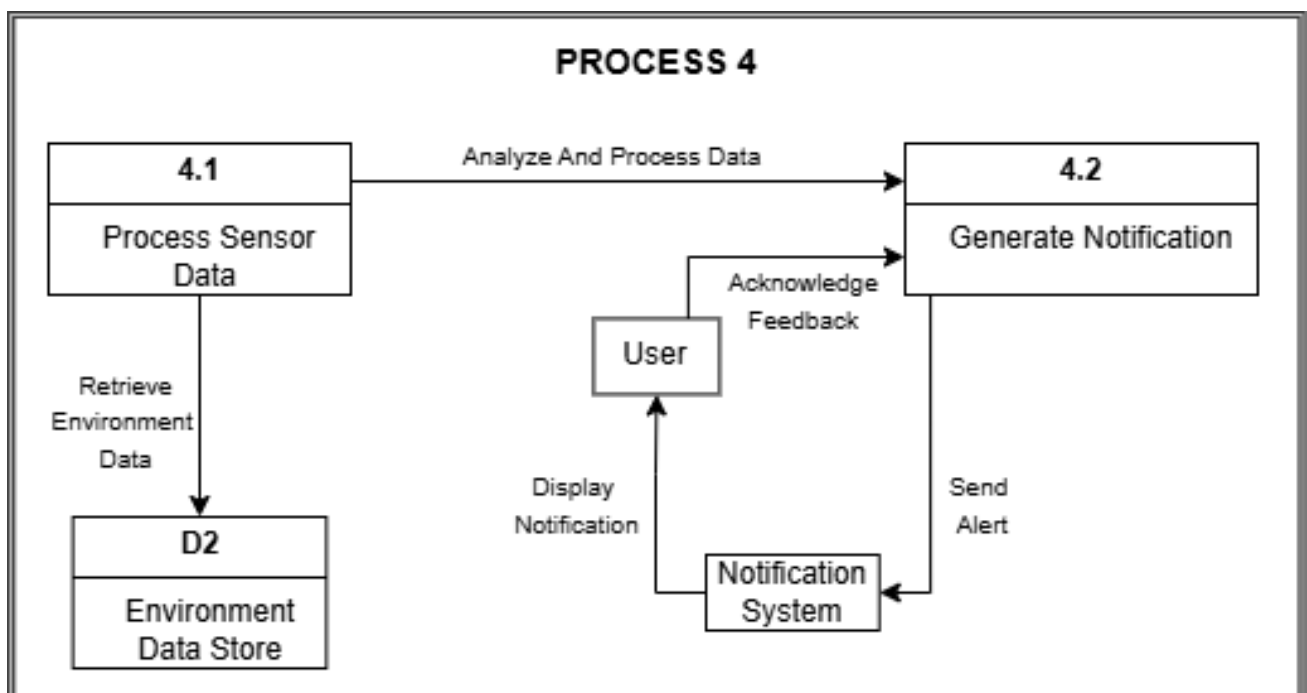
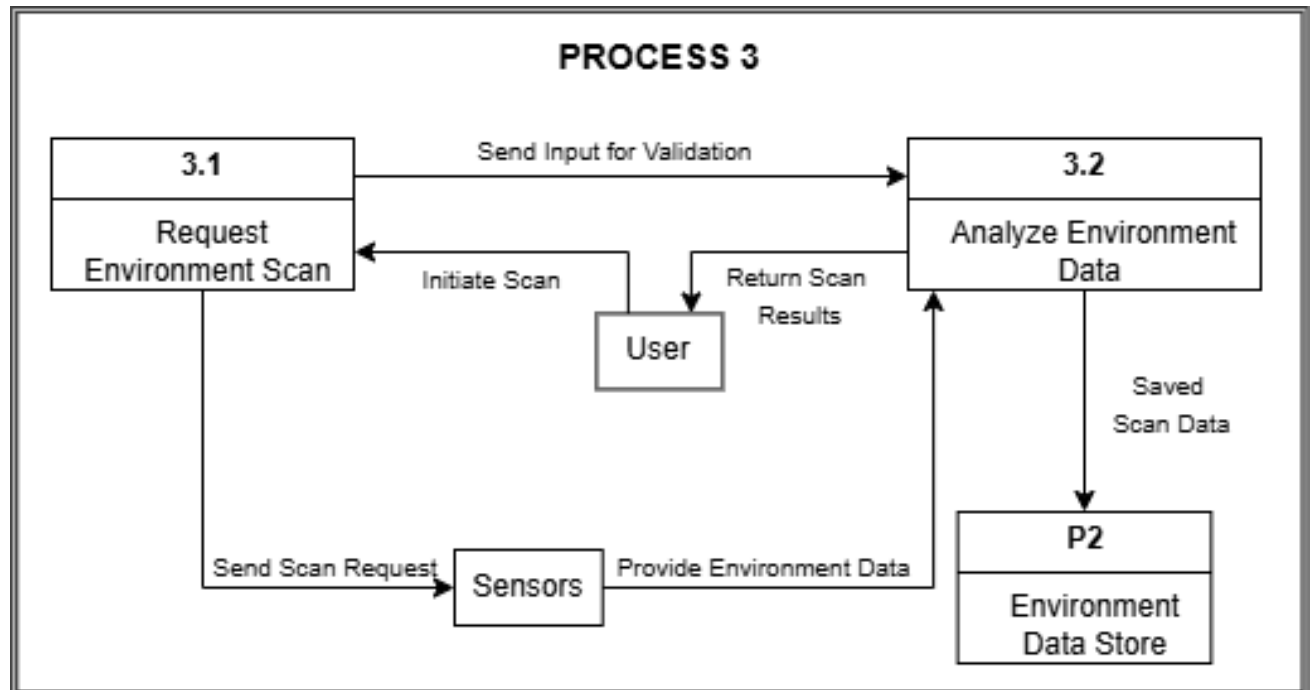
CLASS DIAGRAM



## DATA FLOW DIAGRAM:







## **8. TESTING STRATEGIES:**

### **1. Hardware Testing**

Lens Clarity: Check for distortion and chromatic aberration.

Motion Sensors: Test head tracking accuracy and latency.

Battery & Durability: Assess battery life, charging cycles, and resistance to drops or heat.

Comfort: Evaluate weight distribution and long-term wearability.

### **2. Software Testing**

Compatibility: Test across devices and platforms.

Performance: Check refresh rates, tracking accuracy, and app stability.

Security: Validate data safety and network integrity.

### **3. User Experience Testing**

Immersion: Assess engagement and detect distractions.

Comfort: Ensure usability during extended sessions.

Accessibility: Test features for users with disabilities.

### **4. Functional Testing**

Controllers & Tracking: Verify accuracy and responsiveness.

Peripheral Devices: Ensure compatibility with add-ons like haptic gloves.

### **5. Performance Testing**

Thermal: Monitor heat buildup during use.

Stress Testing: Push limits with resource-intensive apps.

### **6. Compliance**

Health & Safety: Validate against regulations like CE, FCC.

Certifications: Ensure adherence to industry standards.

### **7. User Feedback**

Gather insights from beta testing and improve iteratively.

## **9. REFERENCES:**

### **Books and Articles:**

1. The VR Book by Jason Jerald – An essential guide to designing immersive VR experiences and enhancing usability.

2. Designing Virtual Reality Systems by Gerard Jounghyun Kim – A comprehensive resource on the development and optimization of VR systems.

3. ISO 9241-210 – International standards focusing on ergonomic design for human-system interaction.

4. IEEE 610.12-1990 – Standard definitions and terminology for software engineering and system design.

## **Websites and Online Resources:**

1. Oculus Developer Resources ([developer.oculus.com](https://developer.oculus.com)) – Practical guidelines for VR hardware compatibility, motion tracking, and user interaction design.
2. VRcompare ([vr-compare.com](https://vr-compare.com)) – A platform to compare VR headsets, their specifications, and features.
3. Unity VR Development ([unity.com/learn](https://unity.com/learn)) – Tutorials and documentation for building VR applications using Unity.
4. SteamVR Developer Tools ([steamvr.com](https://steamvr.com)) – Resources for optimizing and developing VR apps for SteamVR-compatible devices.
5. Meta Quest Developer Resources ([meta.com/developers](https://meta.com/developers)) – Official tools and guides for creating apps for Meta Quest VR glasses.
6. Draw.io (<https://app.diagrams.net/>)

## **TOOLS:**

1. Unreal Engine – A powerful tool for designing VR applications using visual scripting (Blueprints).
2. NVIDIA FrameView – A tool for analyzing VR performance and optimizing system resources.
3. OpenXR – A cross-platform API standard for creating VR applications compatible with multiple devices.